

THERMAL POWER PLANT

DCS Operation Notes (Unit-2) - Complete Study Guide

1. Condensate & Feedwater System (Water Path)

Reference: Overview of Condensate System Screen

Ye cycle turbine se nikli hui steam ko wapas pani banakar boiler tak pahunchane ka kaam karti hai.

The Flow Path

Condenser → CEP (A/B) → GSC → LP Heaters (1, 2, 3) → Deaerator → BFP

Key Components & Live Parameters:

- **Condenser:** Steam yahan thandi hokar condensate banti hai. CW flow steam ko thanda karta hai jisse volume kam hota hai aur Vacuum ($\sim 0.140 \text{ kg/cm}^2$) maintain hota hai.
- **CEP (Condensate Extraction Pump):** Hotwell se pani nikal kar pressure ($\sim 21.3 \text{ kg/cm}^2$) se aage bhejta hai. Pump safety ke liye Minimum Recirculation Valve (CD 23) hota hai.
- **LP Heaters (LPH 1, 2, 3):** Turbine ki low-pressure extraction steam se feedwater ko step-by-step pre-heat kiya jata hai. Temperature $\sim 53^\circ\text{C}$ se badhkar $\sim 103^\circ\text{C}$ ho jata hai.
- **Deaerator:** Ek "Open Type" tank jahan steam aur water direct mix hote hain. Pani ko heat karna ($\sim 150^\circ\text{C}$) aur dissolved gases (Oxygen) ko nikalna iska main function hai.
- **DM Make-up:** System mein pani ke loss ko pura karne ke liye DM water hamesha Condenser mein enter karta hai. DM 24 pichle 24 ghante ka system leakage batata hai.

2. Flue Gas Path - Heat Recovery (Before AH)

Reference: Flue Gas Path Before AH Screen

Furnace ki garmi ko alag-alag coils/tubes ke madhyam se steam aur feedwater mein transfer karne ka cycle.

The Flow Path

Furnace → PSH → RH → FSH → LTSH → Economizer (ECO)

Key Components & Live Parameters:

- **Furnace Draft:** Boiler ko hamesha thode negative pressure (-0.624 mmWC) par chalaya jata hai (Balanced Draft) taaki aag bahar na aaye.
- **Superheaters & Reheater:** Flue gas apni heat in tubes ke andar behti steam ko deti hai taaki steam "Superheated" ho jaye.
- **Economizer (ECO):** Boiler drum mein jane se pehle feedwater ko flue gas se aakhiri baar garam kiya jata hai.
- **Temperature Drop:** Shuruat mein gas ka temp ~767°C (PSH) hota hai aur heat exchange ke baad Economizer tak aate-aate ~320°C reh jata hai.

3. Flue Gas Path - Cleaning & Exhaust (After AH)

Reference: Flue Gas Path After AH Screen

Gas ki bachi hui heat ko nikalna aur chimney se bahar chhodne se pehle use saaf karna.

The Flow Path

Air Heater (AH-A/B) → ESP (1, 2, 3, 4) → ID Fans (A/B) → Chimney

Key Components & Live Parameters:

- **Air Heater (AH):** Flue gas apni aakhiri heat fresh hawa (Air) ko de deti hai. Isse boiler efficiency badhti hai aur gas temp ~116°C-119°C tak gir jata hai.
- **O2 Monitoring:** AH inlet/outlet par O2 (Oxygen) level monitor kiya jata hai (~5% - 7%). Ye perfect combustion aur AH leakage ke baare mein batata hai.
- **ESP (Electrostatic Precipitator):** Charon ESP units high voltage ka use karke flue gas se raakh (ash) ko pakad leti hain.
- **ID Fans & Chimney:** ID Fans boiler ka negative draft banate hain aur gas ko khinch kar chimney se nikalte hain. Chimney outlet par SPM, CO, aur NOx limits monitor ki jati hain.

4. Air System (Combustion Support)

Reference: Air & Flue Gas System Screen

Boiler mein aag jalane (combustion) aur coal ko transport karne ke liye hawa ka management.

The Flow Path

Atmosphere → Fans (FD/PA) → SCAPH → AH (Air Heater) → Furnace/Mills

Key Components & Live Parameters:

- **FD Fans (Forced Draft):** Ye bahar se thandi hawa khinchte hain, jo AH mein garam hone ke baad *Secondary Air* ke roop mein furnace mein aag jalane ke kaam aati hai.
- **PA Fans (Primary Air):** Ye hawa AH se guzarkar bahut garam (Hot PA HDR ~320°C) ho jati hai. Iska kaam gile coal ko mills mein sukhana aur udakar burners tak pahunchana hai.
- **SCAPH:** Startup ya low load par jab flue gas thandi hoti hai, tab hawa ko steam se pehle thoda garam kar liya jata hai taaki Air Heater ke metal par acid corrosion na ho.

 **Control Room Tip:** Recirculation valves (jaise CD 23, CD 21) hamesha plant fluctuations ke dauran balance maintain karne aur pumps ko cavitation/overheating se bachane ke liye auto mode mein configure kiye jate hain.